

BEYOND REFERENCE MATCHING: SPECIFICATION-BASED CORRECTNESS EVALUATION FOR DSP IMPLEMENTATIONS

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ABSTRACT

A reference implementation is commonly used as the correctness target for a DSP task. That test is exact when the specification identifies one realization, but a magnitude mask defines a feasible set, not one coefficient vector. We ask when a scalar reference-distance oracle can recover specification membership on an evaluated universe, and give a finite-set exactness criterion in terms of a reference-separability gap. Final labels come from an independent dense-grid verifier, not the construction checker. On 20 mask instances, no coefficient-distance threshold separates independently verified valids from invalids (20/20), including every library choice of reference and finite reference sets of size $K \in \{1, 3, 5, \text{all}\}$. Same-order, same-structure alternatives remain reference-discordant on 20/20 tasks. The gap is positive on eight singleton identities. Reference agreement is therefore a realization diagnostic, not a specification oracle, on these non-unique tasks. Our code is available at: <https://github.com/XianghuiMeng-1020/dsp-semantic-correctness>.

Index Terms— Filter design, specification testing, correctness evaluation, FIR, IIR

1. INTRODUCTION

Evaluating a DSP implementation against a trusted reference is natural when the specification is a unique map: one integer delay, one Fourier convention, one alias-fold period. In those cases, agreement with the reference *is* correctness.

Classical filter design is different. A magnitude mask with a free transition defines a *feasible set* of realizations, not a unique impulse response [1, 2, 3, 4]. Windowed designs [5, 6], frequency sampling [7], equiripple Parks–McClellan approximation [8, 9, 10], and IIR Butterworth or elliptic sections [11, 12, 13, 14] can all meet the same passband, stopband, and stability bounds. That multiplicity is classical and is not our claim.

The evaluation question is sharper. A canonical library call—Hamming `firwin`, or a Butterworth section [15]—

is one occupant of the feasible set. A test that accepts only neighbors of that occupant asks whether a candidate is the *same realization*. It does not ask whether the candidate meets the contract that justified the reference. When can a reference-based scoring rule be a *correct oracle* for a specification-defined DSP task?

We formalize the distinction among a reference realization, a reference acceptance region, and the specification-valid set, and give a finite-universe criterion for the existence of any scalar threshold on distance to that reference. The criterion is not a theorem over all filters: extrema are computed on an explicitly stated evaluated universe after independent verification. Related work treats the mask as a feasible-set design problem [1, 10, 16], uses a golden reference in DSP testing [17], or scores generated code with unique-output unit tests [18, 19]. Those protocols do not decide when a reference ball recovers a non-unique DSP mask.

Contributions. (i) We state soundness, completeness, and exactness of a reference-distance oracle relative to a specification, and a finite-set reference-separability gap. (ii) We separate candidate construction from final labeling with an independent verifier, and probe same-order Type-I feasible sets with predetermined directions. (iii) On 20 mask instances the gap is non-positive in coefficient space for every library reference and for finite multi-reference oracles; it is positive on eight singleton identities.

2. WHEN IS A REFERENCE ORACLE EXACT?

A task t asks for an implementation h (FIR taps or an IIR pair (b, a)). A specification predicate $S_t(h) \in \{0, 1\}$ defines the valid set

$$\mathcal{V}_t = \{h : S_t(h) = 1\}. \quad (1)$$

A reference realization h_r is one distinguished occupant, not the definition of t . Given a distance d and a threshold τ , the reference acceptance region is

$$\mathcal{A}_{\tau,r} = \{h : d(h, h_r) \leq \tau\}. \quad (2)$$

A reference oracle declares “correct” iff $h \in \mathcal{A}_{\tau,r}$. A specification oracle declares “correct” iff $h \in \mathcal{V}_t$.

Soundness is $\mathcal{A}_{\tau,r} \subseteq \mathcal{V}_t$ (no invalid is accepted). *Completeness* is $\mathcal{V}_t \subseteq \mathcal{A}_{\tau,r}$ (no valid is rejected). *Exactness* is $\mathcal{A}_{\tau,r} = \mathcal{V}_t$.

All measurements below are restricted to an explicit finite evaluation universe $\mathcal{U}_t$ (the independently verified candidates for task t). Define

$$D_V(r) = \sup_{h \in \mathcal{V}_t \cap \mathcal{U}_t} d(h, h_r), \quad D_I(r) = \inf_{h \in \mathcal{U}_t \setminus \mathcal{V}_t} d(h, h_r). \quad (3)$$

A scalar threshold classifies every point of $\mathcal{U}_t$ exactly iff there exists τ with $D_V(r) \leq \tau < D_I(r)$. If $D_V(r) \geq D_I(r)$, no threshold on $d(\cdot, h_r)$ recovers specification membership on that universe. The *reference-separability gap*

$$G_r = D_I(r) - D_V(r) \quad (4)$$

is therefore the diagnostic: $G_r > 0$ permits exact threshold separation on $\mathcal{U}_t$; $G_r \leq 0$ is an empirical non-separability certificate on that universe, not a global impossibility theorem. A pair with $S_t(h_{\text{inv}}) = 0$, $S_t(h_{\text{val}}) = 1$, and $d(h_{\text{inv}}, h_r) < d(h_{\text{val}}, h_r)$ is an inversion witness and implies $G_r \leq 0$.

Proposition 1 (finite-universe exactness). On a finite $\mathcal{U}_t$, a scalar threshold on $d(\cdot, h_r)$ classifies every evaluated valid and invalid exactly iff $D_V(r) \leq \tau < D_I(r)$ for some τ . If $G_r \leq 0$, no such τ exists on that universe. The statement is evaluation geometry; it does not certify extrema over all filters.

For a finite reference set $\mathcal{R}_K$ write

$$d_K(h) = \min_{r \in \mathcal{R}_K} d(h, h_r), \quad \mathcal{A}_{\tau, \mathcal{R}_K} = \{h : d_K(h) \leq \tau\}. \quad (5)$$

The same gap applies to d_K .

Two distances are used. The confirmatory coefficient distance d_{coeff} is a relative ℓ_2 after canonicalization (FIR: trailing-zero and Type-I rules; IIR: $a_0 = 1$). Min-length truncation of unequal orders is retained only as a historical baseline. The response distance d_{resp} is the RMSE of $|H|$ on the constrained pass and stop bands. A descriptive operating point $\tau_R = 0.05$ is reported for comparability; it is not the scientific claim.

Fig. 1 states the geometry: the τ -ball is a realization neighborhood; $\mathcal{V}_t$ is the contract.

3. INDEPENDENT VERIFICATION AND PROTOCOL

Construction uses a 4096-point `freqz` search checker. That checker is computationally convenient and is *not* the final labeler. Final VALID/INVALID status is assigned by an independent verifier that does not call the search checker: $N_f = 131072$ magnitude samples, local extremum refinement on each constrained band, second-order-section evaluation for IIR, $a_0 = 1$ normalization, and the registered residual

floors (FIR 10^{-6} , IIR 10^{-3}) and pole-radius bound 0.999. Dense sampling is a numerical certificate, not a continuous-frequency proof. Four previously admitted library `firwin2` occupants on tight low-pass and high-pass masks failed the independent verifier at a constrained-band edge (normalized residual up to 1.6×10^{-2}) and were relabeled invalid. The construction checker had reported a zero residual on the same filters, which is why final labels cannot remain self-certified by the 4096-point search.

Two suites are scored (Table 1). Suite S comprises eight singleton identities (integer delay, circular and linear convolution conventions, lag-0 energy, alias fold, frequency rescale, Nyquist frequency, and $\delta[n - D]$). Suite N comprises 20 instances from one magnitude-mask family: FIR low-pass, high-pass, band-pass, and band-stop at $f_s = 8$ and 16 kHz, each loose and tight (16 FIR), plus IIR low-pass and high-pass at 8 kHz (4 IIR). Tight masks are mechanical (pass ripple and stop ceiling $\times 1/5$; stop edge moved halfway toward the facing pass). Canonical h_r is the shortest-odd Hamming `firwin` (FIR) or the lowest-order `butter` (IIR) that meets the contract.

Suite N candidates include classical library designers, first-principles FIR windowed-sinc and frequency sampling (numpy only) [5, 7], and random-valid search that admits on the search checker only. After independent verification the valid count is 412 (88 library, 24 first-principles, 300 random-valid). Invalids are 144 mechanism mutants plus a second, predetermined near-boundary set (pass-drop and stop-lift at fixed $\varepsilon \in \{0.002, 0.005, 0.010, 0.020\}$), independently verified invalid (160/160 boundary mutants). The primary statistical unit is the specification. Occupant-pooled rates are secondary and are not rates over DSP implementations at large.

For every eligible non-unique FIR task we hold fixed the reference order, length, Type-I symmetry, gain convention, and sampling rate. A length- $2M + 1$ Type-I filter has amplitude $A(\omega) = \theta_M + 2 \sum_{k=1}^M \theta_{M-k} \cos(k\omega)$ and is probed by the linear program $\max q^T \theta$ subject to the pass/stop inequalities on a 512-point-per-band constraint grid, with a predetermined interior margin so vertices survive independent re-verification. Directions (basis, frozen-seed random, tap combinations, library differences) were chosen before observing disagreement. Vertices are independently re-verified. IIR same-order evidence uses independently verified library occupants of equal (b, a) length.

A generated-code arm on the four original loose 8 kHz masks is an existence witness only: no new generations, no model ranking, and no pooling into constructed n .

4. RESULTS

Task-level reference disagreement. On Suite S the descriptive operating point and the gap agree: every independently verified valid matches the canonical output and

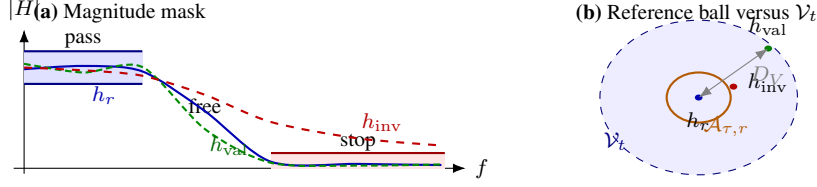


Fig. 1. Same contract, two spaces. (a) Frequency-domain magnitude mask. (b) Schematic realization space: $\mathcal{A}_{\tau,r}$ is a neighborhood of one valid h_r ; $\mathcal{V}_t$ is schematic and is not a Euclidean ball in the same metric. A valid h_{val} may lie outside every useful τ -ball, while a near-boundary invalid h_{inv} may lie closer to h_r (an inversion; $G_r \leq 0$).

Table 1. Study suites after independent verification. Arm G is a witness on four original masks and is not added to constructed n .

Suite	Inst.	Spec. type	Uniq.	Valid	Invalid	Witness
S	8	unique map	yes	12	16	—
N	20	mag. mask	no	412	144	Arm G, 4

Table 2. Finite-universe separability after independent verification. Exact τ means $G_r > 0$ (some threshold classifies all evaluated valids and invalids). FRR is task-macro except the pooled descriptive row.

Diagnostic	Coeff. d	Resp. d	Notes
N tasks with $G_r \leq 0$	20/20	19/20	finite $\mathcal{U}_t$
N inversion witnesses	20/20	—	near-boundary
S tasks with $G_r > 0$	8/8	—	positive control
Task-macro FRR ($\tau_R = 0.05$)	0.900	—	CI [0.871, 0.925]
Pooled descriptive FRR	370/412	—	secondary

$G_r > 0$ on 8/8 tasks (Table 2). The specifications are effectively singleton on the evaluated representation; this is not a uniqueness proof over all functions. On Suite N, 20/20 tasks have at least one independently verified valid with $d_{\text{coeff}}(h, h_r) > \tau_R$. Task-macro FRR is 0.900 (median 0.907, IQR [0.893, 0.947], range [0.727, 0.950]; task-cluster bootstrap 95% CI [0.871, 0.925], $B = 10^4$, seed 20260826). The pooled descriptive rate is $370/412 = 0.898$. Random-valid occupants contribute 298/300 of that pooled count; the task-level 20/20 and the gap results do not depend on how many random samples were kept.

Exact threshold separation. Table 2 reports G_r on the independently verified universe (constructed valids, Type-I probe valids, mechanism invalids, and near-boundary invalids). Coefficient-space $G_r \leq 0$ on 20/20 Suite N tasks: no scalar d_{coeff} threshold has both zero FRR and zero FAR. Response-space $G_r \leq 0$ on 19/20 tasks; one tight IIR instance has a gap of order 10^{-6} . Near-boundary inversion witnesses exist on 20/20 tasks: an independently verified invalid can be closer to h_r than an independently verified valid. The operating point $\tau_R = 0.05$ is therefore not the source of the conclusion. **Same-order, same-structure alternatives.** The Type-I probe produced independently verified, non-trivial, same-order valids that fail the canonical-reference test on 16/16 FIR

Table 3. Confirmatory robustness on Suite N (independently verified labels). Same-order: Type-I probe (FIR) or equal-length library (IIR). Reference choice: any library h_r restores exact d_{coeff} separation. Multi-ref: exact separation under d_K .

Test	Tasks
Same-order valid and reference-discordant	20/20
FIR Type-I probe	16/16
IIR same-order library	4/4
Any library reference exactly separable	0/20
$K = 1, 3, 5$, all-library exact	0/20

tasks, including every tight mask. Trivial encodings (zero-padding, pure sign flip, pure rescaling) were excluded before counting disagreement. All four IIR tasks have independently verified same-order library alternatives that remain reference-discordant. Same-order / same-structure coverage is therefore 20/20 (Table 3). Disagreement is not explained by order change alone.

Reference choice and finite K . Using every independently verified library realization as h_r in turn never restores exact coefficient separation: 0/20 tasks (Table 3). Deterministic multi-reference oracles ($K = 1, 3, 5$, and all library occupants; `firwin/remez/firls/...`, `butter/cheby1/...`) likewise yield 0/20 exact recoveries. A different canonical designer, or a small library of them, does not recover $\mathcal{V}_t \cap \mathcal{U}_t$ on this universe.

Generated-code witness. Among existing generated implementations on the four original loose 8 kHz masks, nine remain independently specification-valid and coefficient-discordant after canonicalization. The event occurs on all four of those instances. This is an existence witness, not a prevalence estimate and not a model comparison.

5. DISCUSSION

A reference-distance oracle answers “near this realization?” It answers “in $\mathcal{V}_t$?” only when $G_r > 0$ on the evaluated universe. That happens on Suite S, where the specification is effectively singleton in the chosen representation. It does not happen on these 20 magnitude-mask instances, for coefficient distance or (with one near-tie) response distance, for any library choice of h_r , or for a finite set of library references.

Response distance is the stronger reference oracle of the two, as classical approximation already suggests: independently verified valids can sit close to h_r on constrained bands while remaining far in coefficient space. It is still not exact on this universe. The single positive response gap (one tight IIR high-pass instance, order 10^{-6}) is a numerical near-tie, not evidence that $|H|$ RMSE recovers S_t in general.

Pooled FRR remains source-dependent: 58/88 library, 14/24 first-principles, and 298/300 random-valid occupants exceed τ_R after independent verification. That is why the primary unit is the task, and why the confirmatory claims are the 20/20 gap, inversion, same-order, and reference-choice counts rather than 370/412.

The study is restricted to one mask family. It is not a claim about beams, adaptive filters, filter banks, or DSP evaluation in general. Requiring Type-I linear phase or a declared maximum order as part of S_t would define a different feasible set. The independent verifier does not certify the continuous mask. IIR same-order evidence uses library occupants rather than a pole/zero feasible-set program. The generated-code arm is a witness on four original masks and is not pooled with the constructed n .

Canonicalization is part of the claim, not a preprocessing footnote. After $a_0 = 1$, Type-I endpoint preservation, and exclusion of zero-padding, sign-flip, and pure-scale encodings, the same-order Type-I alternatives remain distinct transfer functions. The confirmatory distance is therefore not an artifact of comparing unequal truncated coefficient vectors.

A practical reading of Proposition 1 is operational. If a test bench keeps a single golden Hamming or Butterworth design and a scalar coefficient (or band-RMSE) threshold, compute D_V and D_I on the independently labeled universe it actually scores. A negative gap means that no retuning of that threshold can make the reference test exact for that specification. The remedy is to score S_t , or to enlarge the specification until the intended realization is effectively singleton.

Tightening the mask did not restore exactness. The Type-I probe still returned independently verified same-order, reference-discordant valids on all eight tight FIR instances, and coefficient G_r remained negative on all four tight IIR instances. A tighter contract shrinks $\mathcal{V}_t$; it does not, by itself, make $\mathcal{A}_{r,r}$ coincide with that smaller set.

Finite K has the same geometry. The K -oracle replaces one ball by a union of balls around library occupants. On this universe that union still fails to cover the independently verified valids without also covering a nearer invalid, for $K = 1, 3, 5$ and for the full library set. Covering $\mathcal{V}_t \cap \mathcal{U}_t$ would require either the specification predicate or a reference set that is no longer a small canonical catalog.

The four label flips are a protocol result, not a bookkeeping detail. The 4096-point construction checker accepted four tight `firwin2` library occupants that the independent verifier rejects at the band edge after $N_f = 131072$ and local extremum refinement. Those occupants are excluded from the

valid set. Using the construction checker as the final labeler would have kept the older 416 count and would have scored a membership oracle against the same predicate that defined the labels. That circularity is why the authoritative counts are 412 valids and four recorded flips.

The same-order FIR probe is not a search for a convenient counterexample. Directions were frozen (seed 20260826) before disagreement was inspected: basis, frozen-seed random, tap combinations, and predetermined same-order library differences. A Type-I linear program on 512 constraint frequencies per band, with a predetermined interior margin and line-search backoff, produced independently verified alternatives on 16/16 FIR tasks. IIR evidence is weaker by construction: it uses existing same-order library occupants rather than a pole/zero program. Even under that restriction the four IIR tasks remain reference-discordant.

Suite S is the complementary control, not a second leaderboard. On those eight unique-map identities the canonical occupant is effectively the specification in the chosen representation, $G_r > 0$ on 8/8 tasks, and a reference-distance test is exact on the evaluated universe. The contrast is the claim: reference matching is the right oracle when S_t is effectively singleton, and it is not an exact oracle when S_t is a non-unique mask on this universe. A finite catalog of library references remains a catalog of realizations. It does not become $\mathcal{V}_t$ by taking a min-distance union; K equal to the full independently verified library set still fails exact recovery (0/20). The evaluated invalids include 160/160 independently verified near-boundary mutants, so the nearest-invalid side of G_r is not an easy far-field mutant. What $G_r \leq 0$ licenses is narrower than a global impossibility theorem. It is a certificate on the independently labeled $\mathcal{U}_t$ actually scored: no scalar threshold on the chosen d recovers S_t there. It does not prove that every DSP mask is non-separable, that every generated implementation is valid, or that a different distance or a different specification predicate would fail. The task-cluster bootstrap CI [0.871, 0.925] is likewise a statement about these 20 tasks, not about DSP evaluation in general. The four flips sit on tight `firwin2` band edges; they are not random-valid occupants and they are not recycled as mechanism invalids.

6. CONCLUSION

When a DSP specification defines a feasible set, a scalar reference-distance test is an exact correctness oracle on an evaluated universe if and only if the reference-separability gap is positive. On the independently verified Suite N universe that gap is non-positive for coefficient distance on every task, every library reference, and every tested finite K ; same-order feasible-set probing does not remove the mismatch. Reference matching remains the right test for effectively singleton specifications.

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7. REFERENCES

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